

Polynomial Regression, Model Selection, and Regularization

– Lab Assignment Report

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1 Introduction

The goal of this assignment is to build polynomial curve fitting, model selection, and Ridge (L_2) regularization.

What we did on the dataset of 10,000 coordinate pairs (`noisy_3.txt`) are as follows:

1. To understand the **Bias–Variance Tradeoff** across polynomial models of different degree ($d = 1 \dots 10$).
2. Prevented **data leakage** by splitting data into Training (70%), Validation (15%), and Test (15%) sets using fixed random seed.
3. Selected the best polynomial degree based on validation error.
4. Examined how **Ridge (L_2) Regularization** prevents high degree models from overfitting.
5. Measured the underlying random noise in the data and then we verified that our results are stable across different random data splits.

2 Step-by-Step Workflow

The experimental pipeline is organized into the following steps:

1. **Load the Data:** Read those 10,000 points (x_i, y_i) from `input/noisy_3.txt`, where $x \in [-100.0, 100.0]$.
2. **Split the Data (70/15/15):** Partition the data into Training (7,000 points), Validation (1,500 points), and Test (1,500 points) sets using a fixed random seed (`seed = 42`) that is reproducible.
3. **Feature Standardization:** Calculate the mean (μ_{train}) and standard deviation (σ_{train}) using only the training set to prevent data leakage:

$$x_{\text{proc}} = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}} \quad (1)$$

Validation and test inputs are then transformed using these same training parameters.

4. **Build the design matrix $\mathbf{X}$** where each row has columns of $[1, x_i, x_i^2, \dots, x_i^d]$.

5. **Solve for Best Weights:** Compute the exact weights using the closed form Normal Equation:

$$\boldsymbol{\theta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \quad (2)$$

6. **Choose the Best Model:** Compare validation errors for different degrees of $d = 1 \dots 10$ and select $d^* = \arg \min_d \text{MSE}_{\text{val}}(d)$.

7. **Apply Ridge Regularization:** For a high-degree polynomial ($d = 10$), add a penalty matrix $\alpha \mathbf{I}'$ to keep the weights smaller:

$$\boldsymbol{\theta}_{\text{ridge}} = (\mathbf{X}^T \mathbf{X} + \alpha \mathbf{I}')^{-1} \mathbf{X}^T \mathbf{y} \quad (3)$$

where $\mathbf{I}' = \text{diag}(0, 1, 1, \dots, 1)$ leaves the intercept θ_0 unpenalized.

8. **Diagnose the Error:** Check if the leftover errors resemble the Gaussian noise, and test the stability over 5 random splits.

3 Experiments and Results

3.1 Experiment 1: Finding the Best Polynomial Degree ($d = 1 \dots 10$)

We trained polynomials of degrees from 1 to 10 on the train set and evaluated them on all three splits. The results are as shown in Table 1.

Table 1: Model Performance Across Polynomial Degrees $d = 1 \dots 10$

Degree (d)	Train MSE	Val MSE	Test MSE	Train R^2	Val R^2	Test R^2
1	1272.18	1230.45	1297.91	0.0018	0.0022	-0.0034
2	1169.53	1100.43	1154.42	0.0824	0.1076	0.1076
3	1123.06	1067.94	1111.52	0.1188	0.1340	0.1407
4	1103.63	1066.15	1087.53	0.1341	0.1354	0.1593
5	623.73	655.37	612.40	0.5106	0.4685	0.5266
6	395.63	381.16	380.59	0.6896	0.6909	0.7058
7 (Best)	70.11	69.52	69.44	0.9450	0.9436	0.9463
8	70.06	69.68	69.54	0.9450	0.9435	0.9462
9	70.06	69.69	69.58	0.9450	0.9435	0.9462
10	70.05	69.67	69.57	0.9450	0.9435	0.9462

3.1.1 Analysis for different polynomial degree

- **High Underfitting (Degrees 1 to 4):** The models are too simple to follow the waves in the data and the errors are very high ($\approx 1087 - 1298$) and R^2 is near zero.
- **Transition Stage (Degrees 5 and 6):** Here we observe the curves which slowly begin capturing the major data points. The test error drops to 380.59 and R^2 rises to 70.58%.
- **Optimal Model ($d^* = 7$):** The validation error reaches its global minimum at 69.52. On unseen test data, this model achieves $\text{MSE} = 69.44$ and explains 94.63% of all data variation ($R^2 = 0.9463$).
- **Diminishing Returns (Degrees 8 to 10):** The more we increase the degree, the validation error stops increasing whereas the train error is lowered. Thus $d^* = 7$ is selected as the simplest and most accurate model.

3.2 Experiment 2: Visual Comparison of Fitted Curves

Figure 1 shows the fitted polynomial curves of different degree alongside the raw original dataset points.

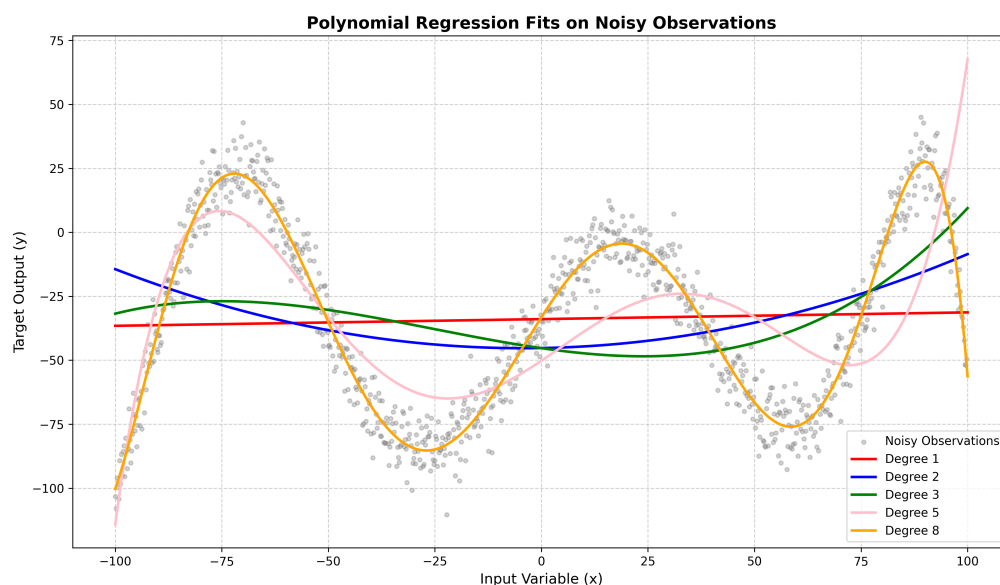


Figure 1: Fitted polynomial curves ($d = 1, 2, 3, 5, 8$) on the noisy dataset points.

Key visual takeaways:

- $d = 1, 2, 3$, these behave like flat lines.
- $d = 5$, this captures the broad pattern but kind of misses sharp turns at the edges.
- $d = 8$ (and $d = 7$), this smoothly cuts through the center of the noisy datas, and it captures the true shape.

3.3 Experiment 3: Bias–Variance Curve and Error Diagnostics

3.3.1 Bias–Variance Tradeoff Curve

Figure 2 shows the training and validation error as degree increases.

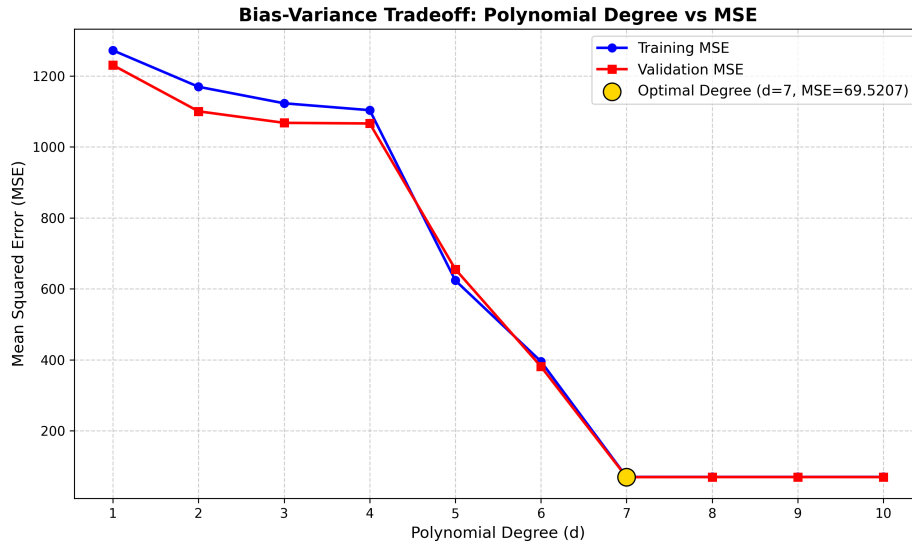


Figure 2: Training MSE and Validation MSE vs. Polynomial Degree ($d = 1 \dots 10$).

The graph shows a sharp drop in error from $d = 1$ to $d = 7$ as bias is eliminated, and then we observe that it is followed by a flat error floor from $d = 7$ onwards.

3.3.2 Residual and Noise Analysis

Figure 3 shows the errors ($e_i = y_i - \hat{y}_i$) produced by the optimal model ($d^* = 7$) on the test set.

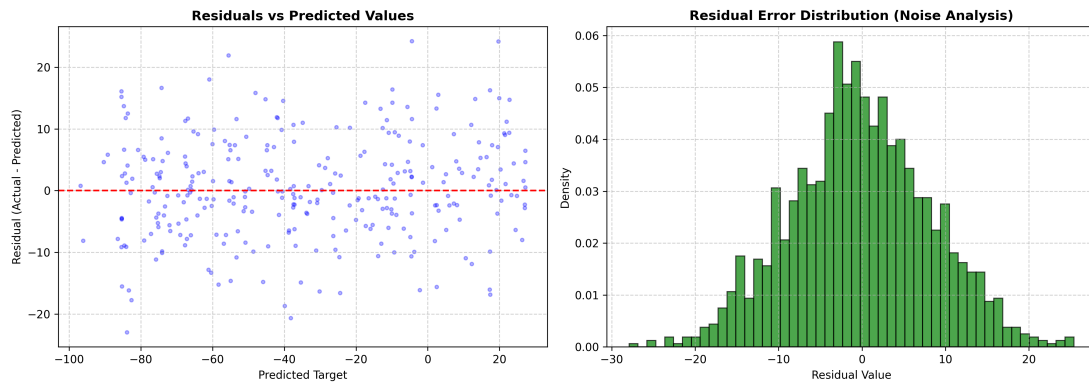


Figure 3: (Left) Residuals vs. Predictions; (Right) Histogram of residual errors.

The error checks confirm the standard regression assumptions:

1. **Uniform Spread (Homoscedasticity):** The noises are evenly spread across all prediction ranges without any narrow or wide patterns.
2. **Zero Average Error:** The average test error is $\bar{e} = -0.19 \approx 0.0$, this proves that the model is unbiased.
3. **Bell-Shaped Curve:** The error histogram in the above graph matches a standard Gaussian normal distribution.
4. **The noise variance:** The measured noise standard deviation is $\sigma_{\text{noise}} = 8.33$. The lowest possible error any model could achieve is:

$$\text{MSE}_{\text{floor}} = \sigma_{\text{noise}}^2 = (8.33)^2 = 69.40 \quad (4)$$

3.4 Experiment 4: Ridge (L_2) Regularization on High-Degree Models ($d = 10$)

To see how the weight penalty controls over-flexibility in a 10th-degree polynomial, we tried testing Ridge regression with penalties $\alpha \in \{0.1, 10.0, 1000.0\}$. The results are summarized in Table 2 and Figure 4.

Table 2: Effect of Ridge Penalty (α) on a Degree 10 Polynomial

Model Configuration	Validation MSE	Test MSE
No Penalty ($\alpha = 0$, Standard $d = 10$)	69.67	69.57
Small Penalty ($\alpha = 0.1$)	69.65	69.54
Medium Penalty ($\alpha = 10.0$)	114.35	115.08
Large Penalty ($\alpha = 1000.0$)	619.91	636.51

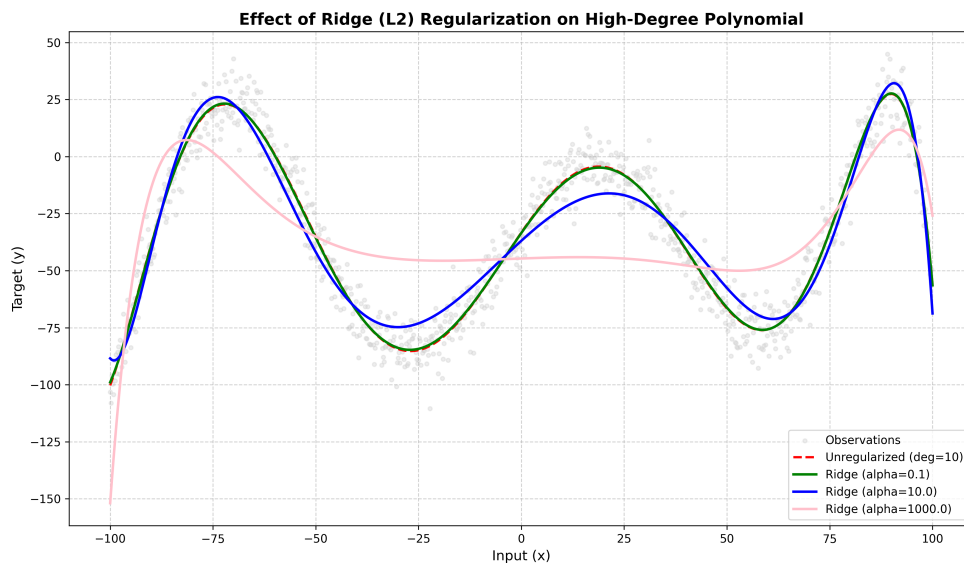


Figure 4: Visual effect of Ridge penalties on a 10th-degree polynomial.

Key takeaways:

- **Small Penalty ($\alpha = 0.1$):** It shrinks the unnecessary high-power terms, lowering test error from 69.57 to 69.54.
- **Excessive Penalty ($\alpha \geq 10.0$):** It over-restricts the curve, which lead to flattening of the function and causing the error to spike ($115.08 \rightarrow 636.51$).

3.5 Experiment 5: Stability Test Across 5 Random Splits

To verify that selecting $d^* = 7$ was consistent for various random split, we experimented with 5 different random split seeds. The results are as shown in Table 3.

Table 3: Stability Evaluation Across 5 Different Random Seeds

Random Seed	Selected Degree (d^*)	Test MSE	Test R^2
42	7	69.44	0.9463
100	7	66.75	0.9470
7	7	69.03	0.9463
2024	7	71.34	0.9453
999	7	70.70	0.9439
Average $\pm$ Std	7	69.45 $\pm$ 1.59	0.9458 $\pm$ 0.0012

From the above table we observe that the validation set has consistently chose $d^* = 7$ across all 5 seeds, which confirms that the model selection is robust.

4 Key Findings

- **The simple models underfit:** Degrees 1 to 4 are too rigid to capture the underlying noises (MSE > 1080).
- **Degree 7 is optimal:** It minimizes the validation error and explains 94.63% of the variance with the fewest unnecessary terms.
- **Error of the noise:** The test error is (69.44).
- **Errors are behaved and evenly spread:** Leftover noises are zero-centered, bell-shaped, and are evenly spread.
- **Ridge regularization helps complex models:** The penalty ($\alpha = 0.1$) prevents the over-fitting in higher degree models, as the capture not only data but also noise.
- **The results are consistent:** Degree 7 was chosen in all 5 independent random data splits (69.45 $\pm$ 1.59) proving that the model is robust.

5 Conclusion

In this assignment, we have successfully done a polynomial curve fitting, model selection, and ridge(L2) regularization. By using 70/15/15 data splitting for train, validation and test and standardizing strictly on the training set, we prevented data leakage. The validation set picked $d^* = 7$ as the optimal model.